



# CYCLISTIC BIKE-SHARE ANALYSIS

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# BUSINESS TASK

Analyze how **Casual Riders** and **Annual Members** use Cyclistic bikes differently to support marketing strategies that increase annual memberships.

# DATA SOURCES AND META DATA

- The Analysis uses **Cyclistic historical Dataset** stored in CSV. The dataset contains ride-level records including timestamps, rider categories station information, and ride duration.
- Initial Data Exploration revealed **several quality issues** such as Zero time duration rides, and invalid timestamps that require cleaning before Analysis.

# DATA CLEANING PROCESS

01



## Convert Timestamps

Converted started\_at and ended\_at to proper datetime format to ensure consistency.

02



## Calculate Ride Duration

Calculated ride\_length as the difference between end time and start time.

03



## Remove Invalid Rides

Removed rides with negative or zero duration and records where start time was greater than end time.

04



## Filter Suspicious Records

Filtered records with missing values, incorrect data or suspicious station information.

05



## Check Duplicates

Checked for duplicate trip\_id records and removed duplicate entries.

06



## Standardize Rider Category

Standardized rider type values to “Member” and “Casual” for uniformity.

07



## Create Time-based Variables

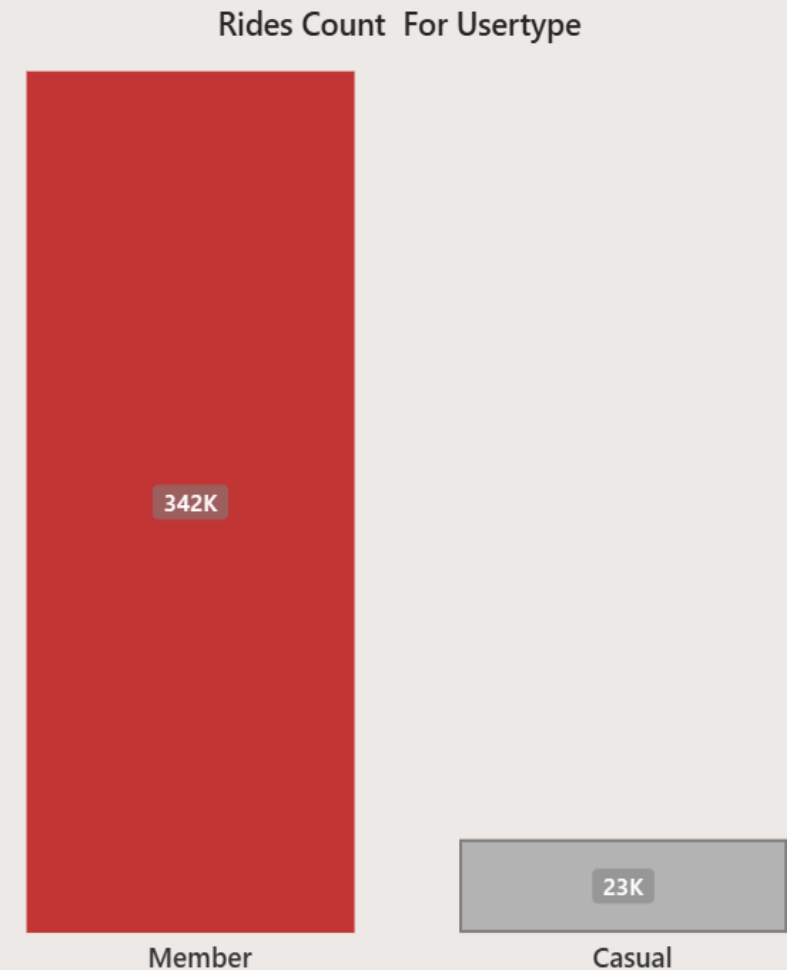
Extracted additional variables such as day\_of\_week, day\_name, month, and hour for analysis.

# SUMMARY OF ANALYSIS

- The Analysis found clear behavioral differences between customer and subscriber groups.
- Members completed significantly **more rides**, whereas casual riders had much longer average ride durations, **indicating recreational usage**.

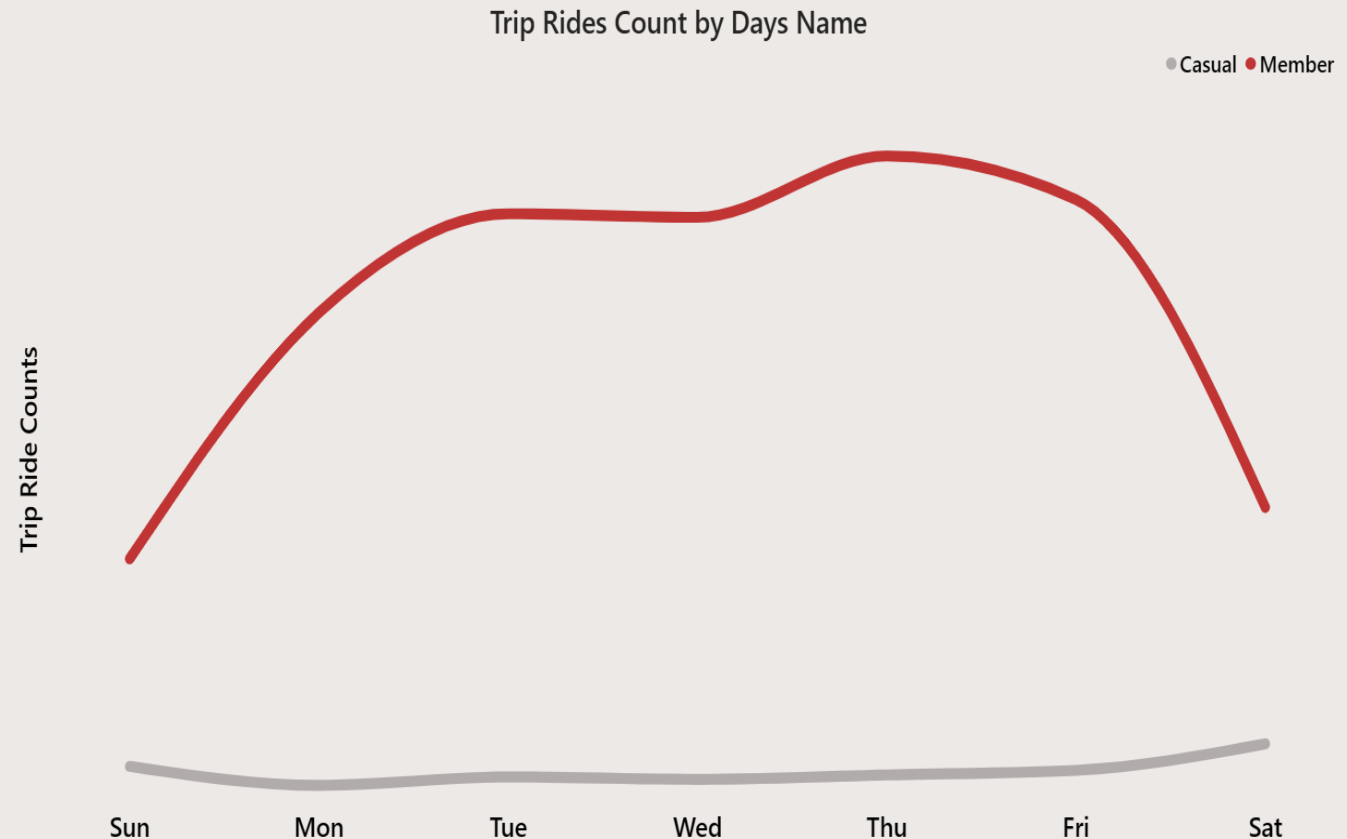
# RIDE COUNT ANALYSIS BY USER TYPE

- ❑ The chart shows that members completed approximately **342K rides**, while casual riders completed only **23K rides**.
- ❑ This indicates that members use Cyclistic bikes far more frequently than casual riders, suggesting **regular commuting behavior**.



# WEEKLY RIDE COUNT COMPARISON

- ❖ The chart shows that members consistently complete more rides throughout the week, **peaking on Thursday** and **least on weekends**.
- ❖ Casual riders record fewer trips overall, with slightly higher activity during weekends, indicating **recreational usage**.



# RECOMMENDATIONS

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## 1. Weekend Membership Campaigns

- Cyclistic should launch special weekend membership promotions and discounted annual plans to encourage casual riders to become members.



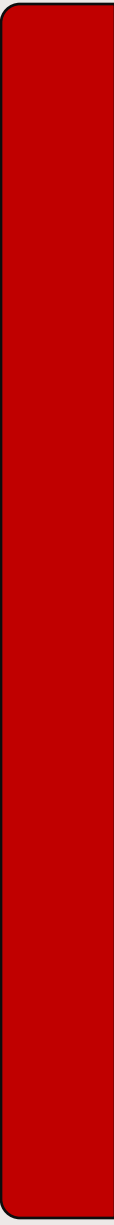
## 2. Promote Cost Savings for Frequent Casual Riders

- Cyclistic should identify frequent casual riders and send personalized messages showing how much money they could save by purchasing an annual membership.



## 3. Digital Marketing Focused on Leisure Riders

- Cyclistic should create targeted social media and mobile app campaigns highlighting the convenience, flexibility, and savings of annual memberships for leisure riders.



# THANK YOU!

Thank you for your attention